Independent review of the two-jump Poisson comparison

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This note proves component estimates for a direct count comparison. It does not prove a new range of the Berry–Esseen inequality. It identifies a false allocation shortcut in the earlier discovery program, gives the correct finite-count variables, and establishes a finite evaluation formula and a useful monotonicity. The accepted small-noise theorems and the local variance-aware moment lemma do not use the false shortcut.

1 Comparison with two Poisson jump sizes

Write $Z_b(V) = b(P_{V/b^2} - V/b^2)$, where $P_a \sim \mathrm{Pois}(a)$. Different occurrences are independent unless stated otherwise; $Z_b(0) = 0$. We use the following published theorem: if independent X_j satisfy $\mathbb{E}X_j \le 0$, $X_j \le b$, and $\sum_j \mathbb{E}X_j^2 \le V$, then

$$\mathbb{E}f\Big(\sum_j X_j\Big) \le \mathbb{E}f(Z_b(V)) \quad \text{for } f \in \mathcal{F}_+^2.$$

Here $\mathcal{F}_+^2$ consists of continuously differentiable functions whose function and derivative are increasing and convex. This is Pinelis, *On the Bennett–Hoeffding inequality*, Proposition 2.8 and Corollary 2.11, printed p. 9, arXiv:0902.4058. In particular $f(x) = (x-y)_+^2$ belongs to this class. The premise is a variance upper bound, not an equality. Zero variance is dealt with directly.

Proposition 1 (Groupwise comparison)**.** *Suppose S_0, S_1 are sums of two independent groups of centered independent variables. The summands in group j are at most $b_j > 0$, and their total variance is at most V_j. Then, for every real y,*

$$\mathbb{E}(S_0 + S_1 - y)_+^2 \le \mathbb{E}\{Z_{b_0}(V_0) + Z_{b_1}(V_1) - y\}_+^2. \tag{1}$$

Consequently, for every $y < b$,

$$\mathbb{P}(S_0 + S_1 \ge b) \le \frac{\mathbb{E}\{Z_{b_0}(V_0) + Z_{b_1}(V_1) - y\}_+^2}{(b-y)^2}. \tag{2}$$

Proof. Condition on $S_1 = z$ and apply the published theorem to the translated hinge $(x+z-y)_+^2$. Integrate this inequality. Then condition on the new independent variable $Z_{b_0}(V_0)$ and replace the second group. All integrands are nonnegative, so Tonelli applies; all expectations are finite by the second-moment bounds. No theorem for a larger class of moment functions is being assumed. Finally $\mathbf{1}_{[b,\infty)}(x) \le (x-y)_+^2/(b-y)^2$. This includes an atom at b and requires the strict inequality $y < b$. □

For the incoming lower tail of W_j, apply this with $S = -W_j$: the jump bound is A on $-U_0$ and $d \ge a_1$ on $-U_1$. The inward bound on U_0 is not a permissible replacement for A.

2 The allocation belongs to the marginal, not to each count

Let $p \in (0,1)$, $q = 1-p$, and let U_0, U_1 be centered conditional laws with variances v_0, v_1. Assume $-A \le U_0 \le a_0$, $-a_1 \le U_1 \le A$, with $a_0, a_1 \ge 0$ and $a_0 + a_1 \le d$. Set

$$s = qv_0 + pv_1 > 0, \quad \lambda = ns, \quad w = \frac{pv_1}{s} \in [0,1], \quad k = np + \delta \in \{0, \dots, n\}.$$

At a legal count $k+i$, the exact group variances are

$$T_{0,i} = \lambda(1-w)\left(1 - \frac{\delta + i}{nq}\right), \qquad T_{1,i} = \lambda w\left(1 + \frac{\delta + i}{np}\right). \tag{3}$$

These follow simply from $v_0 = s(1-w)/q$, $v_1 = sw/p$ and the counts $n-k-i, k+i$. Thus $t_i = \lambda + (\delta + i)(v_1 - v_0)$, but generally $T_{0,i} \ne (1-w)t_i$ and $T_{1,i} \ne wt_i$.

If $p, q \ge 2/5$ and $|\delta| \le D$, then the common inflation

$$T_{0,i} \le \lambda(1-w)\left(1 + \frac{5(D+i)}{2n}\right), \qquad T_{1,i} \le \lambda w\left(1 + \frac{5(D+i)}{2n}\right) \quad (i \ge 1)$$

is valid. The earlier discovery program's incoming inflation therefore has a legitimate interpretation. Its own-count/cubic-repair join did not: there it replaced the two own-count variances by $(1-w)t, wt$ while using w as the marginal allocation in the moment repair.

An exact admissible counterexample to that intermediate bound. Take $n = 42$, $p = 1/2$, $k = 22$, $r = 2/25$, $A = 3/5$, and $\varepsilon = 1/1000$. Let U_0 be the centered two-point law on $\{-A, \varepsilon\}$, with probabilities $\varepsilon/(A+\varepsilon), A/(A+\varepsilon)$, and let $U_1 = -U_0$. This gives an actual four-point marginal $B + U$, not a moment relaxation. Here

$$v_0 = v_1 = v = 3/5000, \quad w = 1/2, \quad \lambda = 63/2500, \quad T_0 = 20v, \quad T_1 = 22v, \quad t = 42v.$$

With $\kappa = -4r$, put $c_0 = A^2 - 4Ar$, $c_1 = A^2 + 4Ar$. The valid bound is $\mathbb{E}W^4 - \kappa\mathbb{E}W^3 \le c_0T_0 + c_1T_1 + 3t^2$. The common-fraction substitute fails by the exact amount

$$\mathbb{E}W^4 - \kappa\mathbb{E}W^3 - \{[c_0(1-w) + c_1w]t + 3t^2\} = \frac{423903}{2500000000} > 0. \tag{4}$$

Indeed $\mathbb{E}U_0^3 = v(\varepsilon - A)$, $\mathbb{E}U_0^4 = v(A^2 - A\varepsilon + \varepsilon^2)$, and independence gives $\mathbb{E}W^4 = 42\mathbb{E}U_0^4 + 3t^2 - 126v^2$. The difference is $v[8rA - 8r\varepsilon - 168A\varepsilon + 42\varepsilon^2]$. The inward sum is $.002 < .072$. The raw threshold at phase $u = -r$ is

$$|H| = \frac{\delta - r}{\sqrt{pq + s}} = \frac{23/25}{\sqrt{.2506}} < 1.84.$$

Thus the counterexample satisfies the intended support and threshold constraints. It invalidates an intermediate discovery inference, not the sharp conjecture, the accepted .03 theorem, or the local moment lemma.

The corrected joint cubic prices. For a quartic minorant $h(x) = (-x^4 + b_3x^3 + b_2x^2 + b_1x + b_0)/M$, $M > 0$, define $\kappa = b_3 - 4r$, $L = b_2 + 3b_3r - 6r^2$, $J = Mh(r)$. Put $\Lambda_0 = \lambda(1-w)$, $\Lambda_1 = \lambda w$, $g_0 = 1 - \delta/(nq)$, $g_1 = 1 + \delta/(np)$, so $t = g_0\Lambda_0 + g_1\Lambda_1$. Let $m > 0$, $C \ge 0$, and assume $a_0 < p$, $a_1 < q$. Write $\mu_j = \mathbb{E}U_j^3$ and $\ell = (pq + s)^{-1/2}$. These support bounds fix the signs of $B + U - p$, giving

$$\rho = \ell^3\{pq(1-2pq) + 3pq(v_0 + v_1) + p\mu_1 - q\mu_0\}.$$

Set $v = pq$ and $\rho_B = (1-2v)/\sqrt{v}$. Choose

$$B \geq \frac{v(1-2v)}{s}\{(1+s/v)^{3/2}-1\},$$

and suppose the resulting lower bound for the entire cubic bracket is nonnegative. If $0 \leq K \leq C\ell^3/n$, one may lower-bound that entire repair using K. For the right-panel bounds, the bracket is positive since $3\min(p,q) - B - d > 0$. Combining the third moments before estimating them then gives the lower bound

$$\begin{aligned} m\mathbb{P}(W > -r) + C(\rho - \rho_B) \;\geq\; & \frac{m}{M}(Lt - 3t^2 + J) + K\lambda\{3p(1-w) + 3qw - B\} \\ & - \Lambda_0 \max_{x\in\{-A,a_0\}} \left\{ \frac{m}{M} g_0(x^2 - \kappa x) + Kx \right\} \\ & - \Lambda_1 \max_{x\in\{-a_1,A\}} \left\{ \frac{m}{M} g_1(x^2 - \kappa x) - Kx \right\}. \end{aligned}$$

To see this, integrate each displayed quadratic against $x^2\, dP_j(x)/v_j$ and use its endpoint maximum; it is convex because $g_j \geq 0$. At $v_j = 0$ the corresponding term is zero directly. This preserves the two physical multipliers. Replacing K in separated signed pieces without first proving the whole bracket nonnegative is not justified. For fixed other parameters the displayed bound is concave in w, owing to $-3t^2$; the endpoint prices are independent of w.

3 Monotonicity removes some allocation work, but not all

Proposition 2 (Variance and allocation monotonicity)**.** *For $0 < d \leq A$, $T \geq 0$, $w \in [0,1]$, and $y \in \mathbb{R}$, set*

$$G(T,w;y) = \mathbb{E}\{Z_A(T(1-w)) + Z_d(Tw) - y\}_+^2.$$

Then G is nondecreasing in T and nonincreasing in w.

Proof. First prove $\mathbb{E}(Z_d(V) - y)_+^2 \leq \mathbb{E}(Z_A(V) - y)_+^2$. For $V > 0$ and integers $N \geq V/d^2$, take independent $B_j \sim \text{Bernoulli}(V/(Nd^2))$. The centered sum $X_N = d\sum_{j=1}^N (B_j - \mathbb{E}B_j)$ has summands bounded above by $d \leq A$ and total variance at most V. The published comparison gives the required bound with X_N in place of $Z_d(V)$. The binomial Poisson limit gives $X_N \Rightarrow Z_d(V)$. Moreover

$$\mathbb{E}X_N^4 \leq d^2 V + 3V^2,$$

since each centered summand has absolute value at most d and independence gives the fourth-moment expansion. Hence the quadratic hinges are uniformly integrable: their tails are controlled by a constant times $(1+|X_N|)^2$, whose squares have uniformly bounded expectation. Truncating a continuous hinge and then removing the truncation justifies the limit. At $V = 0$ the assertion is immediate.

The comparison persists after adding an independent variable with finite second moment: condition on that variable and integrate the translated hinge inequalities. For $w_1 < w_2$, use a common sum consisting of $Z_A(T(1-w_2))$ and $Z_d(Tw_1)$. The remaining variance is $V = T(w_2 - w_1)$, assigned to large jumps at w_1 and small jumps at w_2. The preceding comparison proves monotonicity in w. For $T_1 < T_2$, the law at T_2 is the law at T_1 plus independent centered Poisson increments with variances $(T_2 - T_1)(1-w)$ and $(T_2 - T_1)w$. Conditional Jensen proves monotonicity in T. All assertions include zero intensities. □

For example, choose valid upper group variances $V_0 = \lambda(1-w)\omega_0$, $V_1 = \lambda w\omega_1$ with $0 \leq \omega_0 \leq \omega_1$. Their small-jump fraction is at least w. If $V_0 + V_1 \leq T_+$ and $w \geq w_-$, the incoming hinge is bounded by $G(T_+, w_-; y)$. One must check $\omega_0 \leq \omega_1$; it does not follow from arbitrary count multipliers without a phase condition. On a negative phase, $\delta = \xi + r$, $|\xi| < 1$, so $\delta + i > 0$ for $i \geq 1$ and one may take $\omega_0 = 1$ and a suitable $\omega_1 \geq 1$.

Convexity is false, even for a positive hinge anchor. At $A = 3/5$, $d = 9/125$, $T = 1/40$, $w = 1/4$, and $y = 1/5$,

$$\frac{\partial^2}{\partial w^2} G(T, w; y) = -e^{-815/648} \frac{7873988791147984880 65}{14738261314715557193 3184} < 0. \tag{5}$$

Here is an exact finite verification. Put $\mu = T(1-w)/A^2$, $\nu = Tw/d^2$, $M = A\mu + d\nu$, $z = M + y$. At the stated point $\mu + \nu = 815/648$, $z = 229/720$, and $4d < z < 5d < A$. In a neighborhood, the only atoms below the hinge are zero large jumps and $j = 0, \ldots, 4$ small jumps. Thus

$$G(T, w; y) = T + y^2 - e^{-\mu-\nu} Q(w), \qquad Q(w) = \sum_{j=0}^{4} \frac{\nu^j}{j!} (M + y - jd)^2.$$

Since $(\mu + \nu)' = 385/81$ and $M' = 11/36$, differentiating this finite polynomial shows that the coefficient in (5) is $Q'' - 2(385/81)Q' + (385/81)^2 Q$. The independent rational checker reconstructs it. Therefore a concave own-count bound minus this hinge need not be concave. Allocation endpoints alone cannot be inferred from convexity. Monotone bounds on closed allocation intervals remain valid.

4 The complete hinge is a finite complement

For $A = 3/5$, $d = 9/125$, set $a = 3/125$, so $A = 25a$, $d = 3a$. Let $S = 25P_\mu + 3P_\nu$. For every real K,

$$\mathbb{E}(S - K)_+^2 = 625\mu + 9\nu + (25\mu + 3\nu - K)^2 - e^{-\mu-\nu} \sum_{25j+3l \le K} \frac{\mu^j \nu^l}{j!l!} (K - 25j - 3l)^2. \tag{6}$$

The sum is finite (empty if $K < 0$). This is the identity $x_+^2 = x^2 - (-x)_+^2$, followed by the exact first two moments of S. It includes all upper-tail mass and every lattice collision; there is no truncation error. The physical anchor is $K = (y + A\mu + d\nu)/a$, and the physical numerator is a^2 times (6). A rational K is allowed.

The independent implementation uses the probability-generating function $e^{\mu z^{25} + \nu z^3} = \sum_{j \ge 0} c_j z^j$ instead of a double sum:

$$c_0 = 1, \qquad j c_j = 25\mu c_{j-25} + 3\nu c_{j-3}, \quad c_j = 0 \ (j < 0).$$

The finite deficit is $\sum_{j \le K} (K - j)^2 c_j$. For rational $x = \mu + \nu \ge 0$, let $S_N = \sum_{j=0}^{N} x^j/j!$ and $N + 2 > x$. The positive exponential series gives

$$S_N \le e^x \le S_N + \frac{x^{N+1}}{(N+1)!} \frac{1}{1 - x/(N+2)}.$$

Reciprocals enclose e^{-x}. A lower exponential bound gives an upper hinge numerator because the deficit is subtracted. This direction is essential. It remains valid at zero intensity and at integer K.

For an alternative box bound with fixed K, the uncentered numerator $\mathbb{E}(S - K)_+^2$ increases with each intensity by ordinary Poisson coupling. Use upper intensity endpoints in that numerator and a positive lower bound on $b/a + 25\mu + 3\nu - K$ in the denominator. A centered physical anchor generally moves K with the intensities; in that case one should apply Proposition 2, or account for this movement explicitly.

5 Bounding all remaining incoming counts

Every legal count has variance at most $n\max(v_0, v_1) \le (5/2)\lambda$ when $p, q \ge 2/5$. Put $\mu = (5/2)\lambda/A^2$ and assume $0 < A < 1$, $r \ge 0$. For an integer $i \ge 2$, the Poisson hinge at integer anchor i gives

$$\mathbb{P}(W_{k+i} \le -i - r) \le \frac{\mu^{i+1}}{(i+1)!D_i^2}, \qquad D_i = i(1/A - 1) + r/A + \mu > 0.$$

Indeed $(j-i)_+^2 \le \binom{j}{i+1}$ for integers $j \ge 0$ and $i \ge 2$. For $j = i + l$, $l \ge 1$, the ratio $\binom{i+l}{i+1}/l^2$ starts at one and is nondecreasing, since

$$\frac{\binom{i+l+1}{i+1}/(l+1)^2}{\binom{i+l}{i+1}/l^2} = \frac{l(i+l+1)}{(l+1)^2} \ge 1 \quad \Longleftrightarrow \quad (i-1)l \ge 1.$$

Poisson factorial moments give the numerator. Since D_i increases, for any $m \ge 2$ with $\mu < m + 2$ the complete tail is bounded by

$$\sum_{i=m}^{n-k} \mathbb{P}(W_{k+i} \le -i - r) \le \frac{\mu^{m+1}}{(m+1)!D_m^2} \frac{1}{1 - \mu/(m+2)}.$$

The added terms beyond legal counts are nonnegative numbers in the majorant; no nonexistent conditional law is invoked. The condition $i \ge 2$ matters: at $i = 1, j = 3$, the proposed pointwise bound would read $4 \le 3$. At $\lambda = 0$ handle the deterministic conditional noises directly.

Status. These are complete component derivations under the stated hypotheses. The published Poisson comparison remains an external theorem. Exact checks cover the transport counterexample, a genuine failure of convexity, finite-complement arithmetic, actual conditional convolutions and rejected shortcuts. They are not a closed-domain count cover, nor an independent proof of the published theorem. No old certificate or manuscript file is changed.